Supplementary Material
Augmented quantization: mixture models for risk-oriented sensitivity analysis

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Abstract

This document provides the supplementary material of the article *Augmented quantization: mixture models for risk-oriented sensitivity analysis*. It contains additional numerical results, the proof of Proposition 3, and details on the *FindR* step.

1. Augmented Quantization without perturbation

Here we perform AQ without the clusters perturbation, on a one-dimensional sample $S_u = (\mathbf{X}_i)_{i=1}^{300} \in \mathbb{R}^{300}$ to represent the uniform mixture of density $f = \frac{1}{3}\mathbb{1}_{[0,1]} + \frac{2}{3}\frac{\mathbb{1}_{[0.3,0.6]}}{0.3}$. Here, $R_U(a,b)$ denotes the uniform measure on the interval $[a,b]$. We start from two representatives $R_1 = R_U(0,0.5)$ and $R_2 = R_U(0.5,1)$, and Figure 1 shows the clusters obtained after each of the first 9 iterations. The algorithm is rapidly stuck in a configuration far from the optimal quantization, with representatives around $R_U(0.26,0.93)$ and $R_U(0.19,0.63)$.

The obtained quantization error here is 4.0×10^{-2}, almost 10 times higher than the optimal quantization error obtained with Augmented Quantization including clusters perturbation, equal to 4.4×10^{-3}.

2. Proof of proposition 3

This document provides the proof of the Proposition 3 of the associated manuscript, that shows the asymptotic clustering consistency:

Proposition 3 (Asymptotic clustering consistency.)**.** *If $R_j \in \mathcal{R}_s$ for $j \in \mathcal{J}$ and $\mathbf{X}_i \sim R_{J_i},\ i = 1,\dots,n$ with $(J_i)_{i=1}^n$ i.i.d. sample with same distribution as J, then*

$$\lim_{n,N\to+\infty} \mathbb{E}\left(\mathcal{E}_p(\boldsymbol{C}^\star(\boldsymbol{R},J,n,N))\right) = 0.$$

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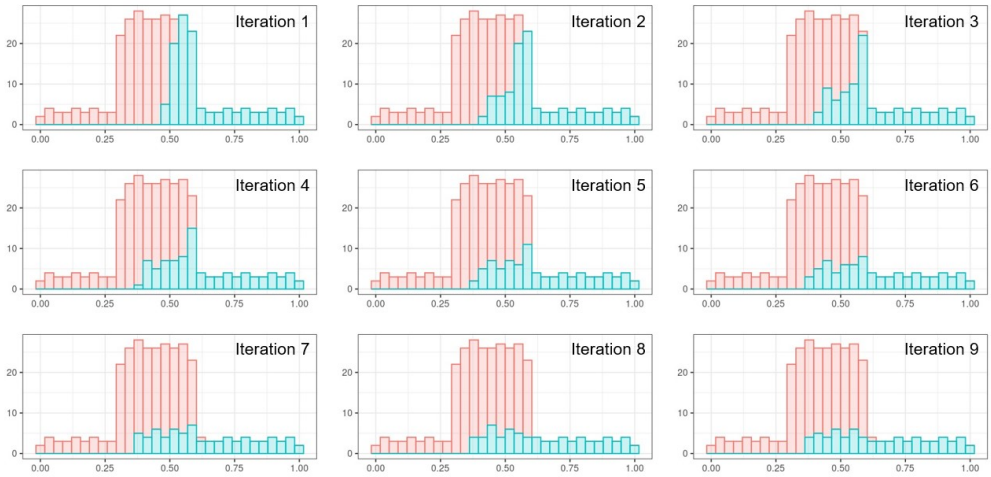


Figure 1: Distribution of clusters 1 and 2 in the 1D uniform case when performing the method without the clusters perturbation.

We recall and complete our notations used to discuss clustering:

- The representative mixtures are $(R_1, \dots, R_\ell) \in \mathcal{R}_s^\ell$
- J is a random variable with $\forall j \in \mathcal{J},\ P(J = j) = p_j$
- $(\mathbf{X}_k)_{k=1}^n$ are i.i.d. data samples and $\mathbf{X}_k$ has probability measure R_J
- $(J_i)_{i=1}^N$ are i.i.d. samples with same distribution as J
- $(\mathbf{Y}_i)_{i=1}^N$ are i.i.d. mixture samples and $\mathbf{Y}_i$ has probability measure R_{J_i}
- The index of the representative whose sample is closest to $\mathbf{x}$, $I_N(\mathbf{x}) = \underset{i=1,\dots,N}{\arg\min} \|\mathbf{x} - \mathbf{Y}_i\|$
- The mixture sample the closest to $\mathbf{x}$, $a_N(\mathbf{x}) = \mathbf{Y}_{I_N(\mathbf{x})}$
$$= \arg \min_{\mathbf{y} \in \{\mathbf{Y}_1, \dots, \mathbf{Y}_N\}} \|\mathbf{x} - \mathbf{y}\|$$
- The k-th data sample that is assigned to cluster j, $\tilde{\mathbf{X}}_{k,N}^{(j)} = \mathbf{X}_k \mid J_{I_N(\mathbf{X}_k)} = j$. This of course does not mean that the k-th sample will belong to cluster j, which may or may not happen. It is a random vector whose probability to belong to cluster j will be discussed.

The space of measures of the representatives is $\mathcal{R}_s := \{\beta_c R_c + \beta_{\text{disc}} R_{\text{disc}}, R_c \in \mathcal{R}_c, R_{\text{disc}} \in \mathcal{R}_{\text{disc}}, \beta_c + \beta_{\text{disc}} = 1\}$ where

- $\mathcal{R}_c$ is the set of the probability measures R_c for which the associated density (i.e. their Radon–Nikodým derivative with respect to the Lebesgue measure), exists, has finite support and is continuous almost everywhere in $\mathbb{R}^m$,

- $\mathcal{R}_{\text{disc}}$ is the set of the probability measures of the form $R_{\text{disc}} = \sum_{k=1}^{t} c_k \delta_{\gamma_k}$, with δ_{γ_k} the Dirac measure at γ_k and $c_k \in]0, 1]$ such that $\sum_{k=1}^{t} c_k = 1$.

As $\mathcal{R}_s$ is stable under linear combinations, it follows that $R_J \in \mathcal{R}_s$, and it can be written $R_J = \beta_c R_c + \beta_{\text{disc}} \sum_{k=1}^{t} c_k \delta_{\gamma_k}$. We denote by f_c the probability density function (PDF) associated with R_c, and we assume that $\beta_{\text{disc}} > 0$ and $\beta_c > 0$, since the reasoning is simpler when either of them is zero.

2.1. Convergence in law of the mixture samples designated by the clustering

2.1.1. Convergence in law of $a_N(\mathbf{X}_k)$

The law of the closest sample from the mixture to the k-th data sample is similar, when N grows, to that of any representative, for example the first one.

Proposition 4. *$\forall k = 1, \dots, n,\ a_N(\mathbf{X}_k)$ converges in law to $\mathbf{Y}_1$:*

$$a_N(\mathbf{X}_k) \xrightarrow{\mathcal{L}} \mathbf{Y}_1$$

Proof. Let $\boldsymbol{\alpha} \in \mathbb{R}^m$. We denote $A(\boldsymbol{\alpha}) = \{\mathbf{y} \in \mathbb{R}^m, \forall i = 1, \dots, m,\ y_i \leq \alpha_i\}$. The cumulated density function (CDF) of $a_N(\mathbf{X}_k)$ is

$$\begin{aligned} &F_{a_N(\mathbf{X}_k)}(\boldsymbol{\alpha}) = \\ &\beta_c \int_{\mathbb{R}^m} P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \mathbf{x}) f_c(\mathbf{x}) d\mathbf{x} \\ &+ \beta_{\text{disc}} \sum_{r=1}^{t} c_r P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \gamma_r)\ . \end{aligned}$$

The proof works by showing that the CDF of $a_N(\mathbf{X}_k)$ tends to that of $\mathbf{Y}_1$ when N grows.

Limit of $\beta_{\text{disc}} \sum_{r=1}^{t} c_r P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \gamma_r)$.
If $\gamma_r \in A(\boldsymbol{\alpha})$, then $P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \gamma_r) \geq P(\exists i \in \{1, \dots, N\}, \mathbf{Y}_i = \gamma_r)$. As a result, $P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X} = \gamma_r) \underset{N \to +\infty}{\longrightarrow} 1$.
Else, if $\gamma_r \notin A(\boldsymbol{\alpha})$, then $P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X} = \gamma_r) \leq P(\forall i \in \{1, \dots, N\}, \mathbf{Y}_i \neq \gamma_r)$. Consequently, $P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X} = \gamma_k) \underset{N \to +\infty}{\longrightarrow} 0$.
Finally, it follows that $\beta_{\text{disc}} \sum_{r=1}^{m} c_r P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X} = \gamma_r) \underset{N \to +\infty}{\longrightarrow} \beta_{\text{disc}} \sum_{r\ s.t\ \gamma_r \in A(\boldsymbol{\alpha})} c_r$

Limit of $\beta_c \int_{\mathbb{R}^m} P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \mathbf{x}) f_c(x) dx$.

Let $\mathbf{x} \in \mathbb{R}^m$ such that f_c is continuous at $\mathbf{x}$ and $\mathbf{x} \notin \{\mathbf{x} \in \mathbb{R}^m, \exists i \in \{1, \dots, d\},\ x_i = \alpha_i\}$.

- If $f_c(\mathbf{x}) = 0$ then $\forall N, P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \mathbf{x}) f_c(\mathbf{x}) = 0$

- Case when $\mathbf{x} \in A(\boldsymbol{\alpha})$ and $f_c(\mathbf{x}) > 0$. Then, $\exists(\varepsilon_1, \ldots, \varepsilon_m) \in (\mathbb{R}_+^\star)^m, x_k = \alpha_k - \varepsilon_k$.

  By continuity, $\exists \eta \in]0, \underset{1 \leq k \leq m}{\arg\min}\, \varepsilon^k[, \forall \mathbf{x}' \in \mathcal{B}(\mathbf{x}, \eta), f_c(\mathbf{x}') > 0$, with $\mathcal{B}(\mathbf{x}, \eta)$ the ball of radius η and center $\mathbf{x}$. It leads to

$$P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \mathbf{x}) \geq P(\exists i \in \{1, \ldots, N\}, \mathbf{Y}_i \in \mathcal{B}(\mathbf{x}, \eta)) \underset{N \to +\infty}{\longrightarrow} 1$$

- Case when $\mathbf{x} \notin A(\boldsymbol{\alpha})$ and $f_c(\mathbf{x}) > 0$. Then, $\exists \varepsilon > 0, \mathbf{x}_k = \alpha_k + \varepsilon$. By continuity, $\exists \eta \in]0, \varepsilon[, \forall \mathbf{x}' \in \mathcal{B}(\mathbf{x}, \eta), f_c(\mathbf{x}') > 0$, with $\mathcal{B}(\mathbf{x}, \eta)$ the ball of radius η and center $\mathbf{x}$. Then,

$$P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \mathbf{x}) f_c(\mathbf{x}) \leq P(\forall i \in \{1, \ldots, N\}, \mathbf{Y}_i \notin \mathcal{B}(\mathbf{x}, \eta)) \underset{N \to +\infty}{\longrightarrow} 0.$$

We then have, $P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \mathbf{x}) f_c(\mathbf{x}) \underset{N \to +\infty}{\overset{a.e.}{\longrightarrow}} \mathbb{1}_{A(\boldsymbol{\alpha})}(\mathbf{x}) f_c(\mathbf{x})$.

The Dominated Convergence Theorem gives,

$$\beta_c \int_{\mathbb{R}^m} P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid \mathbf{X}_k = \mathbf{x}) f_c(\mathbf{x}) d\mathbf{x} \underset{N \to +\infty}{\longrightarrow} \beta_c \int_{A(\boldsymbol{\alpha})} f_c(\mathbf{x}) d\mathbf{x}$$

Limit of $F_{a_N(\mathbf{X}_k)}(\boldsymbol{\alpha})$.

Finally, we have,

$$F_{a_N(\mathbf{X}_k)}(\boldsymbol{\alpha}) \underset{N \to +\infty}{\longrightarrow} F_Y(\boldsymbol{\alpha}) \,.$$

□

2.1.2. Convergence in law of $a_N(\mathbf{X}_k) \mid J_{I_N(\mathbf{X}_k)} = j$

The random vector $a_N(\mathbf{X}_k) \mid J_{I_N(\mathbf{X}_k)} = j$, i.e., the closest mixture sample to the k-th data sample, when constrained to belong to the j-th cluster and when N grows, follows the distribution of the j-th representative R_j.

Proposition 5.

$$\forall j \in \mathcal{J},\ a_N(\mathbf{X}_k) \mid J_{I_N(\mathbf{X}_k)} = j \xrightarrow{\mathcal{L}} \mathbf{Y}_1 \mid J_1 = j$$

Proof. As in the previous Proposition, $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_m) \in \mathbb{R}^m$ and $A(\boldsymbol{\alpha}) = \{\mathbf{x} \in \mathbb{R}^m, \forall i = 1, \ldots, m,\ x_i \leq \alpha_i\}$.

To shorten the notations, we write the k-th sample point associated to the j-th cluster $\tilde{\mathbf{X}}_{k,N}^{(j)} \coloneqq \mathbf{X}_k \mid J_{I_N(\mathbf{X}_k)} = j$.

$$
\begin{aligned}
&F_{a_N(\tilde{\mathbf{X}}_{k,N}^{(j)})}(\boldsymbol{\alpha}) \\
&= P(a_N(\tilde{\mathbf{X}}_{k,N}^{(j)}) \in A(\boldsymbol{\alpha})) \\
&= P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}) \mid J_{I_N(\mathbf{X}_k)} = j) \\
&= \sum_{i=1}^{N} P(I_N(\mathbf{X}_k) = i, \mathbf{Y}_i \in A(\boldsymbol{\alpha}) \mid J_{I_N(\mathbf{X}_k)} = j) \\
&= NP(I_N(\mathbf{X}_k) = 1, \mathbf{Y}_1 \in A(\boldsymbol{\alpha}) \mid J_{I_N(\mathbf{X}_k)} = j)
\end{aligned}
$$

Bayes' Theorem gives,

$$
\begin{aligned}
&P(I_N(\mathbf{X}_k) = 1, \mathbf{Y}_1 \in A(\boldsymbol{\alpha}) \mid J_{I_N(\mathbf{X}_k)} = j) \\
&= \frac{P(J_{I_N(\mathbf{X}_k)} = j \mid I_N(\mathbf{X}_k) = 1, \mathbf{Y}_1 \in A(\boldsymbol{\alpha}))}{P(J_{I_N(\mathbf{X}_k)} = j)} \\
&\quad \times P(I_N(\mathbf{X}_k) = 1, \mathbf{Y}_1 \in A(\boldsymbol{\alpha})) \\
&= \frac{P(J_1 = j \mid \mathbf{Y}_1 \in A(\boldsymbol{\alpha}))P(I_N(\mathbf{X}_k) = 1, \mathbf{Y}_1 \in A(\boldsymbol{\alpha}))}{P(J_{I_N(\mathbf{X}_k)} = j)} \\
&= \frac{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}) \mid J_1 = j)P(J_1 = j)}{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}))} \\
&\quad \times \frac{P(I_N(\mathbf{X}_k) = 1, \mathbf{Y}_1 \in A(\boldsymbol{\alpha}))}{P(J_{I_N(\mathbf{X}_k)} = j)} \\
&= \frac{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}) \mid J_1 = j)p_j}{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}))} \frac{P(I_N(\mathbf{X}_k) = 1, \mathbf{Y}_1 \in A(\boldsymbol{\alpha}))}{p_j} \\
&= \frac{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}) \mid J_1 = j)P(I_N(\mathbf{X}_k) = 1, \mathbf{Y}_1 \in A(\boldsymbol{\alpha}))}{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}))}
\end{aligned}
$$

Then

$$
\begin{aligned}
&F_{a_N(\tilde{\mathbf{X}}_{k,N}^{(j)})}(\boldsymbol{\alpha}) \\
&= N \frac{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}) \mid J_1 = j)P(I_N(\mathbf{X}_k) = 1, \mathbf{Y}_1 \in A(\boldsymbol{\alpha}))}{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}))} \\
&= \frac{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}) \mid J_1 = j)}{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}))} \sum_{i=1}^{N} P(I_N(\mathbf{X}_k) = i, \mathbf{Y}_i \in A(\boldsymbol{\alpha})) \\
&= P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}) \mid J_1 = j) \frac{P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}))}{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}))}
\end{aligned}
$$

Proposition 4 gives $\frac{P(a_N(\mathbf{X}_k) \in A(\boldsymbol{\alpha}))}{P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}))} \underset{N \to +\infty}{\longrightarrow} 1$.

As a result,

$$
F_{a_N(\tilde{\mathbf{X}}_{k,N}^{(j)})}(\boldsymbol{\alpha}) \underset{N \to +\infty}{\longrightarrow} P(\mathbf{Y}_1 \in A(\boldsymbol{\alpha}) \mid J_1 = j) \, .
$$

□

2.2. Wasserstein convergence to the representatives

2.2.1. Wasserstein convergence of $a_N(\tilde{\mathbf{X}}_{k,N}^{(j)})$

The mixture sample that is both the closest to $\mathbf{X}_k$ and coming from R_j has a distribution that converges, in terms of Wasserstein metric, to R_j. This is a corollary of Proposition 5.

Corollary 1.

$$\mathcal{W}_p(\mu_N^j, R_j) \underset{N\to+\infty}{\longrightarrow} 0,$$

where μ_N^j is the measure associated to $a_N(\tilde{\mathbf{X}}_{k,N}^{(j)})$.

Proof. As stated in [**?**], convergence in p-Wasserstein distance is equivalent to convergence in law plus convergence of the pth moment. As the distributions of $\mathcal{R}$ have bounded supports, $X \xrightarrow{\mathcal{L}} Y \Longrightarrow E(\| X \|^p) = E(\| Y \|^p)$. □

2.2.2. Wasserstein convergence of $(a_N(\tilde{\mathbf{X}}_{k,N}^{(j)}))_{k=1}^n$

The above Wasserstein convergence to the j-th representative for a given data sample $\mathbf{X}_k$ also works on the average of the $\mathbf{X}_k$'s as their number n grows.

Proposition 6.

$$\lim_{n,N\to+\infty} \mathbb{E}(\mathcal{W}_p(\mu_{n,N}^j, R_j)) = 0,$$

where $\mu_{n,N}^j$ is the empirical measure associated to $(a_N(\tilde{\mathbf{X}}_{k,N}^{(j)}))_{k=1}^n$.

Proof. By the triangle inequality applied to the $\mathcal{W}_p(,)$ metric,

$$\begin{aligned}\mathbb{E}(\mathcal{W}_p(\mu_{n,N}^j, R_j)) \leq & \mathbb{E}(\mathcal{W}_p(\mu_{n,N}^j, \mu_N^j)) + \\ & \mathbb{E}(\mathcal{W}_p(\mu_N^j, R_j)) \,.\end{aligned}$$

As the distributions of $\mathcal{R}_s$ have a bounded support, then for all $q \in \mathbb{N}$,
$\exists M_1, \forall N \in \mathbb{N}, \int_{\mathcal{X}} \| \mathbf{x} \|^q \, \mu_N^j(d\mathbf{x}) \leq M_1$.
Then, as shown in [**?**],

$$\exists M_2, \forall N \in \mathbb{N},\ \mathbb{E}(\mathcal{W}_p(\mu_{n,N}^j, \mu_N^j)) \leq M_2 G(n)$$

where $G(n) \underset{n\to+\infty}{\longrightarrow} 0$.
Let $\varepsilon > 0$,

$$\exists n_0, \forall n > n_0, \forall N,\ \mathbb{E}(\mathcal{W}_p(\mu_{n,N}^j, \mu_N^j)) < \frac{\varepsilon}{2} \,,$$

and Corollary 1 gives

$$\exists N_0, \forall N > N_0,\ \mathbb{E}(\mathcal{W}_p(\mu_{N,j}, R_j)) < \frac{\varepsilon}{2} \,.$$

Finally,

$$\forall n, N > max(n_0, N_0),\ \mathbb{E}(\mathcal{W}_p(\mu^j_{n,N}, R_j) < \varepsilon\ .$$

□

2.2.3. Wasserstein convergence of $C^\star_j(\boldsymbol{R}, J, n, N)$ to R_j

The j-th cluster produced by the *FindC* algorithm, $C^\star_j(\boldsymbol{R}, J, n, N)$, can be seen as the i.i.d. sample $(\tilde{\mathbf{X}}^{(j)}_{k,N})_{k=1}^{n_j}$, with $\sum_{j=1}^{\ell} n_j = n$. We have $n_j \to +\infty$ when $n \to +\infty$, provided $p_j > 0$.

We denote $\nu^j_{n_j,N}$ the empirical measure associated to $(\tilde{\mathbf{X}}^{(j)}_{k,N})_{k=1}^{n_j}$.

The triangle inequality implies that

$$\begin{aligned}\mathbb{E}(\mathcal{W}_p(\nu^j_{n_j,N}, R_j)) \leq & \mathbb{E}(\mathcal{W}_p(\nu^j_{n_j,N}, \mu^j_{n_j,N}))+ \\ & \mathbb{E}(\mathcal{W}_p(\nu^j_{n_j,N}, R_j))\ .\end{aligned}$$

Furthermore,

$$\begin{aligned}&\mathbb{E}(\mathcal{W}_p(\nu^j_{n_j,N}, \mu^j_{n_j,N})) \\ &\leq \mathbb{E}(\frac{1}{n_j}\sum_{k=1}^{n_j} ||\tilde{\mathbf{X}}^{(j)}_{k,N} - a_N(\tilde{\mathbf{X}}^{(j)}_{k,N})||^p) \\ &\leq \mathbb{E}(||\tilde{\mathbf{X}}^{(j)}_{1,N} - a_N(\tilde{\mathbf{X}}^{(j)}_{1,N})||^p) \\ &\leq \mathbb{E}(||\mathbf{X}_1 - a_N(\mathbf{X}_1)||^p |\ J_{I_N(\mathbf{X}_1)} = j)\end{aligned}$$

The law of total expectation gives

$$\begin{aligned}&\mathbb{E}(||\mathbf{X}_1 - a_N(\mathbf{X}_1)||^p) \\ &\quad = \sum_{j=1}^{\ell} p_j \mathbb{E}(||\mathbf{X}_1 - a_N(\mathbf{X}_1)||^p |\ J_{I_N(\mathbf{X}_1)} = j)\ .\end{aligned}$$

Since $a_N(\mathbf{X}_k)$ is the closest sample in $\{\mathbf{Y}_1, \ldots, \mathbf{Y}_N\}$ to $\mathbf{X}_k$, we have

$$\begin{aligned}&\mathbb{E}(||\mathbf{X}_1 - a_N(\mathbf{X}_1)||^p) \\ &= \mathbb{E}(\frac{1}{N}\sum_{k=1}^{N} ||\mathbf{X}_k - a_N(\mathbf{X}_k)||^p) \\ &\leq \mathbb{E}(\mathcal{W}_p((\mathbf{X}_k)_{k=1}^N, (\mathbf{Y}_i)_{i=1}^N)) \underset{N\to+\infty}{\longrightarrow} 0\end{aligned}$$

The last two relations give,

$$\begin{aligned}&\forall\, j \in \mathcal{J},\\ &\quad \mathbb{E}(||\mathbf{X}_1 - a_N(\mathbf{X}_1)||^p |\ J_{I_N(\mathbf{X}_1)} = j) \underset{N\to+\infty}{\longrightarrow} 0\end{aligned}$$

Therefore,

$$\begin{cases} \lim\limits_{n_j,N\to+\infty} \mathbb{E}(\mathcal{W}_p(\nu^j_{n_j,N}, \mu^j_{n_j,N})) = 0 \,, \\ \lim\limits_{n_j,N\to+\infty} \mathbb{E}(\mathcal{W}_p(\nu^j_{n_j,N}, R_j)) = 0 \text{ by Proposition 6.} \end{cases}$$

2.3. Convergence of the expected quantization error

We have

$$\begin{aligned} &\mathcal{E}_p(\boldsymbol{C}^\star(\boldsymbol{R}, J, n, N)) \\ &= \left(\sum_{j\in\mathcal{J}} p_j w_p(C^\star_j(\boldsymbol{R}, J, n, N))^p \right)^{1/p} \\ &\leq \sum_{j\in\mathcal{J}} p_j w_p(C^\star_j(\boldsymbol{R}, J, n, N)) \end{aligned}$$

where $w_p(C_j) := \mathcal{W}_p(C_j, R^\star_j(C_j)) = \min\limits_{r\in\mathcal{R}} \mathcal{W}_p(C_j, r)$, denotes the local error associated for a given cluster C_j. By definition of the local error, for $j \in \mathcal{J}$, $w_p(C^\star_j(\boldsymbol{R}, J, n, N)) \leq \mathcal{W}_p(C^\star_j(\boldsymbol{R}, J, n, N), R_j)$. As just seen in Section 2.2.3,

$$\begin{aligned} &\lim_{n_j,N\to+\infty} \mathbb{E}\left(\mathcal{W}_p(\nu^j_{n_j,N}, R_j)\right) \\ &= \lim_{n,N\to+\infty} \mathbb{E}\left(\mathcal{W}_p(C^\star_j(\boldsymbol{R}, J, n, N), R_j)\right) \\ &= 0 \,. \end{aligned}$$

Then,

$$\lim_{n,N\to+\infty} \mathbb{E}\left(w_p(C^\star_j(\boldsymbol{R}, J, n, N))\right) = 0 \,,$$

and finally,

$$\lim_{n,N\to+\infty} \mathbb{E}\left(\mathcal{E}_p(\boldsymbol{C}^\star(\boldsymbol{R}, J, n, N))\right) = 0 \,.$$

3. *FindR* step for mixtures of Dirac and uniform measures

In this section we provide details about *FindR* step with the family $\mathcal{R}$ introduced in Section 4. This step consists of finding, in the sense of Proposition 2, the optimal parameters

$$(\alpha_1, \ldots, \alpha_m, a_1, \ldots, a_m, \sigma_1, \ldots, \sigma_m)$$

with α_k a boolean $\sigma_k = 0$ if and only if $\alpha_k = 0$.. The optimization is performed separately on each marginal, to avoid the computation of multivariate Wasserstein distances. Then, for each cluster C_j, we define here its k-th marginal: $C^k_j = \{x_k \,:\, (x_1, \ldots, x_m) \in C_j\}$, and we minimize

$$\mathcal{W}_2\left(C^k_j, R_k(\alpha_k, a_k, \sigma_k)\right),$$

where

- R_k is a uniform between $a_k - \frac{\sigma_k}{2}$ and $a_k + \frac{\sigma_k}{2}$ if the boolean $\alpha_k = 1$, with $\sigma_k \in \{0.25, 0.5, 0.75, 1\}$.
- If $\alpha_k = 0$, R_k is a Dirac at $a_k \in [0, 1]$, and $\sigma_k = 0$.

Then, the four possibilities are tested for the pair (α_k, σ_k), and we have the following analytical result:

- If $\alpha_k = 0$, then $\sigma_k = 0$ and R_k is a Dirac measure at a_k. The optimal value for a_k is

$$a_k^\star = \frac{1}{\text{card}(C_j)} \sum_{x \in C_j^k} x$$

  the barycenter of C_j^k.

- If $\alpha_k = 1$, then $\sigma_k \in \{0.25, 0.5, 0.75, 1\}$ and R_k is a 1D-uniform measure with support center a_k and support width σ_k. The optimal value for a_k is

$$a_k^\star = \int_0^1 Q_j^k(q)(-6q + 4)dq,$$

  with Q_j^k the empirical quantile function of C_j^k.

As specified in the manuscript, the finite candidate set for the width of the uniform distribution is a deliberate choice to facilitate discrimination between variables. However, analytical results could also be derived for the optimization of the parameter pair $(\alpha_k, \sigma_k) \in \mathbb{R}^2$, allowing for a fully continuous parametric family $\mathcal{R}$.

4. Screening on G-function

We consider a test case based on the G-function introduced in [**?**], defined as

$$G(\mathbf{X}) = \prod_{i=1}^{d} \frac{|4X_i - 2| + a_i}{1 + a_i},$$

where the input variables $X_i \sim \mathcal{U}(0, 1)$ are independent, $a_i \geq 0$ are coefficients controlling variable influence, and d is the dimension. In this study, we set $d = 10$.

The coefficients a_i determine the impact of each variable on the output: smaller values of a_i correspond to stronger influence, whereas larger a_i indicate weaker influence. The objective is to perform screening and identify the most influential variables on the risk event $G(\mathbf{X}) > q_{95}$, where q_{95} is the 0.95-quantile of $G(\mathbf{X})$.. We do this using AQ by considering a mixture of distributions with independent marginals, where each marginal is either a Dirac delta or a uniform distribution with support width 1.

For this example, we choose:

- $a_i = 99$ for $i \in \{1, 2, 4, 5, 6, 9, 10\}$, making these variables essentially non-influential,
- $a_3 = 0$, giving X_3 the largest influence,

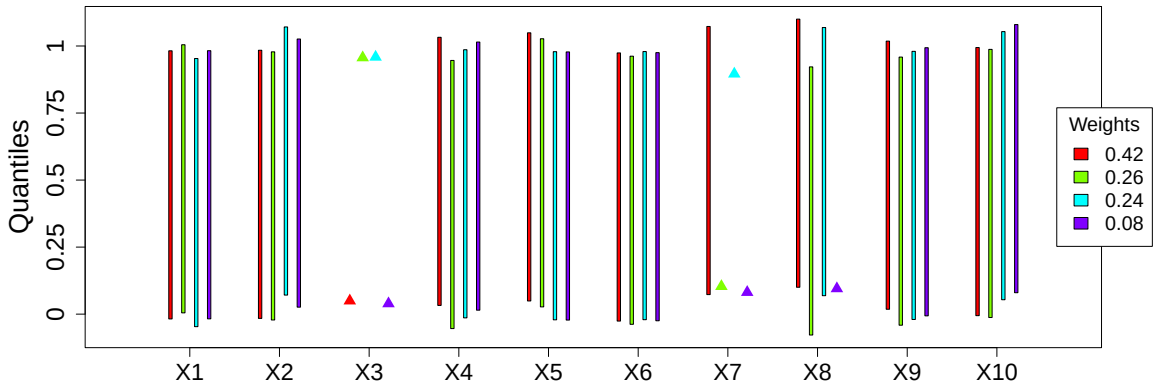


Figure 2: Scenarios leading to $G(\mathbf{X}) > q_{95}$ in the flooding test case: mixture of four distributions with Dirac and large uniform independent marginals. Each distribution is associated with a color (red, green, blue and purple). The mixture weights are 0.42 for the red component, 0.26 for the green component, 0.24 for the blue component, and 0.08 for the purple one, as indicated on the right side of the plot. A vertical bar represents a uniform distribution, while a triangle marks the location of a Dirac.

- $a_7 = 2$, giving X_7 a moderate influence,
- $a_8 = 4$, giving X_8 a small influence.

From independent samples $(\mathbf{X}_i)_{i=1}^{6000}$, we obtain $n = 281$ realizations of

$$(F_1(X_1), \ldots, F_{10}(X_{10})) \,\Big|\, G(\mathbf{X}) > q_{95}.$$

Figure 2 provides the obtained mixture, that confirms the expectations given the chosen parametrization:

- For all variables $i \notin \{3, 7, 8\}$, the obtained mixture exhibits only large uniform representatives, indicating that these variables are non-influential, as expected.
- For X_3, only Dirac representatives are identified, showing its strong influence on $G(\mathbf{X}) > q_{95}$.
- Variable X_7 can take any value in 42% of the scenarios (red representative), while only specific values of this variable lead to $Y > q_{95}$ in the remaining cases (Dirac for the three other representatives). This confirms that its influence is smaller than that of X_3, but still significant.
- The influence of X_8 is highlighted in only 8% of the cases (purple scenario), confirming that although it still has an effect, this effect is smaller than for X_3 and X_7.

This analysis is entirely confirmed by the target HSIC indices, provided in Figure 3.

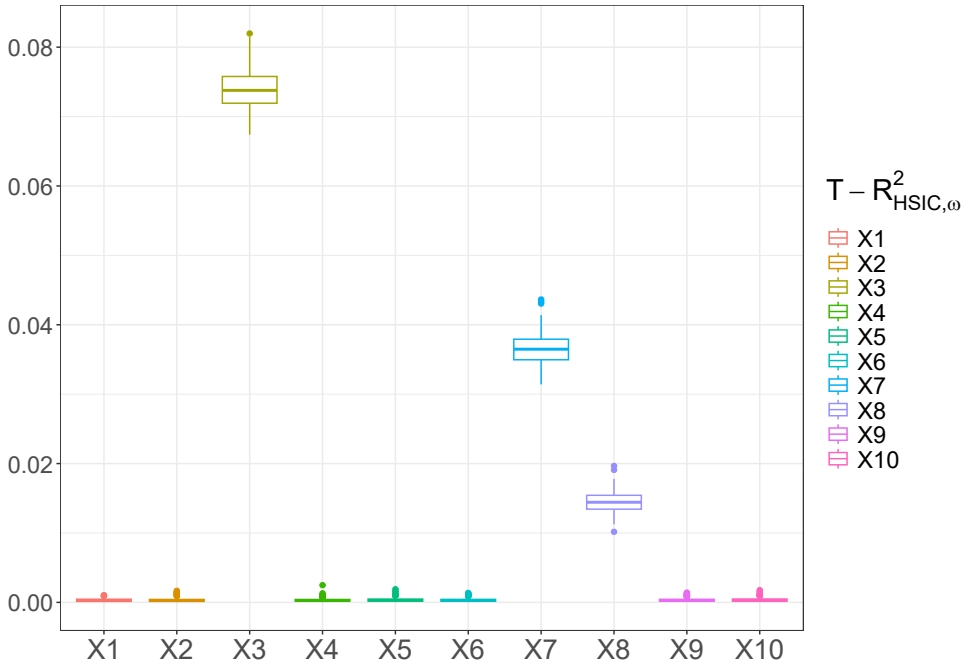


Figure 3: Target HSIC T-$R^2_{\text{HSIC},\omega}(X_i, G(\mathbf{X}))$ estimated from samples of size 6000 in the flooding test case, plotted on a logarithmic scale.